

AP CSP | Lesson 13: Counted Loops

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: C

C is correct: evaluate the old value 4 plus 2, then store 6. A confuses the update amount with the result. B ignores assignment. D doubles x instead of adding 2.

Q2. Answer: A

A matches the counted-loop structure. B ignores the block boundaries. C counts outputs instead of iterations. D changes the computation rather than repeating the statements.

Q3. Answer: D

D is correct: count becomes 2, 4, then 6. A performs one iteration. B counts iterations but ignores adding 2. C has an incorrect initial value or arithmetic.

Q4. Answer: B

B gives an initial sum without erasing earlier contributions. A initializes too late. C repeatedly discards the running total. D reads an uninitialized total.

Q5. Answer: A and C

A follows zero-iteration semantics. C counts 2×3 executions. B incorrectly excludes zero. D adds the two loop counts instead of applying the inner count for each outer iteration. Award one mark only for A and C together.

Q6. Answer: C

C displays the current value before each update. A reverses the statement order. B ignores saved updates. D adds an extra output after the loop that was not specified.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: Outputs: 2, 5, 8. Final count: 11.

1 mark: Gives first output 2.

1 mark: Gives second output 5.

1 mark: Gives third output 8.

1 mark: Gives final count 11, distinct from the displayed outputs.

Q8. 4 marks

Answer: After iteration 1: 8. After iteration 2: 20. After iteration 3: 25. Final output: 25.

1 mark: Gives first running total 8.

1 mark: Gives second running total 20.

1 mark: Gives third running total 25.

1 mark: Gives final displayed value 25 once after the loop.

Q9. 4 marks

Answer: (a) 5. (b) 4. (c) count $\leftarrow$ 0 before the loop. (d) Resetting each time discards previous increments, so reset-then-increment would leave 1.

1 mark: Gives actual final count 5.

1 mark: States exactly four iterations execute.

1 mark: Initializes count to 0 before the loop.

1 mark: Explains that resetting inside each iteration erases earlier increments; may also state final count 1 for reset then increment.

Q10. 4 marks

Answer: (a) 6. (b) 2. (c) Tick, Tick, Tick, Done, Tick, Tick, Tick, Done. (d) Done is outside the inner loop but inside the outer loop, so it executes after each complete group of three Tick outputs.

1 mark: Gives six Tick outputs.

1 mark: Gives two Done outputs.

1 mark: Gives the complete order as two groups of three Tick followed by one Done.

1 mark: Explains that Done lies outside the inner loop and executes only after that inner loop finishes, once per outer iteration.

Q11. 4 marks

Answer: (a) total = 6; i = 4. (b) total = 9, because updating i first makes the additions 2, 3 and 4.

1 mark: Gives original final total 6.

1 mark: Gives original final i 4.

1 mark: Gives swapped final total 9.

1 mark: Explains that swapped order adds the updated values 2, 3 and 4.

Q12. 4 marks

Answer: total $\leftarrow$ 0

REPEAT 2 TIMES

```
{
  minutes  $\leftarrow$  INPUT()
  total  $\leftarrow$  total + minutes
}
```

average $\leftarrow$ total / 2

DISPLAY(average)

Tests: 30, 50 $\rightarrow$ 40; 0, 0 $\rightarrow$ 0.

1 mark: Initializes a total to 0 before the loop.

1 mark: Uses REPEAT 2 TIMES with an INPUT and an addition to the running total inside the body.

1 mark: Calculates total / 2 and displays that average once after the loop.

1 mark: Gives both test outputs correctly: 40 for 30 and 50; 0 for 0 and 0.